Principles of Robot Autonomy I

Practice Questions

Problem 1. Coordinate Rotations and Jacobians

A mobile robot moves in the plane. Its body frame B has its x-axis pointing forward, and its heading θ is measured counterclockwise from world frame W. The velocity expressed in the body frame is

$$\mathbf{v}^B = \begin{bmatrix} v \\ 0 \end{bmatrix}.$$

The robot has state $\mathbf{x} = [p_x, p_y, \theta]^T$ and control $\mathbf{u} = [v, \omega]^T$, where $\dot{\theta} = \omega$.

(a) Write the rotation matrix $R_{WB}(\theta)$ that maps a vector expressed in B to its coordinates in W. Use it to express the full state dynamics $\dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{u})$.

(b) Derive the state and control Jacobians A and B, evaluated at $\theta^* = \pi/2$, $v^* = 2$ m/s. Provide both the continuous and discrete forms.

(c) At this operating point, explain the effect of a small positive heading error on the robot's world-frame x-velocity.

Problem 2. LQR

For the same linear system $\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u}$ and the same initial state, two infinite-horizon LQR designs use the cost

$$J = \int_0^\infty \left(\mathbf{x}^T Q \mathbf{x} + \mathbf{u}^T R \mathbf{u}\right) dt.$$

Controller 1 uses Q_1, R_1. Controller 2 uses $Q_2 = 3Q_1$ and $R_2 = 3R_1$. Assume a LQR feedback law exists, and there are no input constraints.

Which statements must be true? Select all that apply and briefly justify.

☐ A. Both designs produce the same optimal feedback gain K.

☐ B. Controller 2 must use smaller control inputs because R_2 is larger.

☐ C. For any fixed state and input trajectories, the cost evaluated with Q_2, R_2 is three times the cost evaluated with Q_1, R_1.

☐ D. The optimal closed-loop state trajectory must change because Q and R have changed.

Justification:

Problem 3. ROS

Below is a stop-sign controller we have implemented. Assume that all necessary imports, package configurations, and executable permissions are correct.

```python
class StopSignController(Node):
    def __init__(self) -> None:
        super().__init__("stop_sign_controller")

        self.stop_detected = False

        self.control_pub = self.create_publisher(
            Twist, "/cmd_vel", 10
        )

        self.stop_sub = self.create_subscription(
            Bool,
            "/stop_sign",
            self.stop_callback,
            10
        )

        self.control_timer = self.create_timer(
            0.1, self.control_callback
        )

    def stop_callback(self, msg: Bool) -> None:
        if msg.data:
            self.stop_detected = True
            self.get_logger().info("Stop sign detected!")

    def control_callback(self) -> None:
        cmd = Twist()
        if self.stop_detected:
            cmd.linear.x = 0.0
            cmd.angular.z = 0.0
        else:
            cmd.linear.x = 1.0

        self.control_pub.publish(cmd)

if __name__ == "__main__":
    rclpy.init()
    node = StopSignController()
    rclpy.spin(node)
    rclpy.shutdown()

```

(a) Which topics are the node subscribed to and publishing to? How often does the StopSignController publish to this topic?

(b) Suppose the node publishing to `\stop_sign` unexpectedly stops running. Does the StopSignController continue publishing velocity commands? Describe its behavior in the following cases:

  (a) A stop sign has never been detected.

  (b) A stop sign was previously detected.

(c) A student claims that the `control_callback()` function only executes when a new message arrives on `/stop_sign`. Is this correct? Briefly explain.

Problem 4. Collision-Free Path Planning

A circular mobile robot with radius $r = 0.30$ m navigates an environment containing obstacles. Two straight corridors have widths 0.50 m and 0.80 m, respectively.

A path planner represents the robot as a point located at its center.

Which of the following statements are correct? Select all that apply and briefly justify.

- ☐ A. Inflating every obstacle by the robot's radius allows the planner to treat the robot as a point while accounting for its physical size.
- ☐ B. The robot can safely traverse the 0.50 m corridor if it follows the exact centerline.
- ☐ C. The 0.80 m corridor is geometrically traversable by the robot.
- ☐ D. Increasing the resolution of the occupancy grid will make the 0.50 m corridor traversable.
- ☐ E. If the robot's center follows a path through unoccupied grid cells, the robot is guaranteed to avoid collisions without any additional collision checking or obstacle inflation.

Justification: